AI Tools for Learning & Education

How to use LLMs and AI tools to learn better, not less

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The AI Education Explosion

85% of teachers and **86%** of students

used AI during the 2024–2025 school year

All three major LLMs launched education-specific modes within weeks of each other:

ChatGPT	Claude	Gemini
Study Mode	Learning Mode	Guided Learning
July 2025	August 2025	August 2025

The question isn't *whether* to use AI — it's **how to use it well**.

The Core Tension

AI can give you any answer instantly

Copy question → paste answer
Zero effort, zero learning

your choice →

AI can help you understand deeply

Guide, quiz, explain, challenge
Real learning, lasting knowledge

"AI should make students think harder, not less." — Anthropic

What We'll Cover

1. **The Big Three**
   - ChatGPT Study Mode
   - Claude Learning Mode
   - Gemini Guided Learning
   - Side-by-side comparison
2. **Specialized Tools**
   - NotebookLM
   - Perplexity AI
   - Khanmigo
   - Math, coding, writing tools
3. **How to Learn**
   - The right way vs. wrong way
   - Prompting for learning
   - Study session workflow
4. **Pitfalls & Tips**
   - Hallucinations
   - Academic integrity
   - Privacy & data
   - Free access guide

ChatGPT Study Mode

Launched: July 2025 | **Access:** Free tier (select "study and learn" in tools)

How it works:

- Guides you step-by-step instead of giving answers
- **Refuses** direct answers until you engage with the material
- Asks questions to test your understanding
- Calibrates to your skill level

Interactive Learning:

- 70+ math & science topics
- Pythagorean theorem, ideal gas law, lens equations...
- Visualizations for complex concepts

Example interaction

You: How do I find the derivative of $f(x) = x^3 + 2x$?

ChatGPT: Before I show you, let me ask -- do you remember the power rule? What happens when you differentiate x^n?

You: nx^{n-1}?

ChatGPT: Exactly! Now try applying that to each term...

ChatGPT Study Mode: Features

Upcoming features

- Clearer visualizations for complex concepts
- Goal setting & progress tracking
- Deeper personalization by skill level

Also useful for learning:

- **Canvas** — side-by-side editing for writing/code
- **Memory** — remembers your course, level, preferences
- **Custom GPTs** — course-specific chatbots
- **Voice mode** — conversational tutoring
- **File upload** — problem sets, diagrams

Demo idea

Open ChatGPT → "study and learn" → try a math problem. Notice how it guides rather than solves.

ChatGPT Edu:

- Dedicated tier for universities
- GPT-4o access, enterprise-grade privacy
- Admin controls, SSO, group workspaces

Claude: Learning & Explanatory Modes

Launched: August 2025 | **Stat:** 76% of users reported improved understanding

Learning Mode

Socratic pair-programming

- Pauses and inserts `#TODO` for you to write code
- You complete 5–10 lines yourself
- Claude reviews your work and guides further
- Learn by doing, not by reading

Explanatory Mode

Senior dev mentoring

- Narrates every decision as it works
- Explains architecture choices, tradeoffs
- Shares best practices in real-time
- Like looking "under the hood"

Claude for Education (since April 2025): deployed at Northeastern (50K users), LSE, Champlain College. *"AI should make students think harder, not less."*

Claude: Example Interaction

Learning Mode example

```
You:  Help me write a function
that reverses a linked list.

Claude:  Let's build this together.
I'll set up the structure --
you write the core logic.

def reverse_list(head):
  prev = None
  current = head
  # TODO: Write a while loop that
  # iterates through the list and
  # reverses the pointers.
  # (about 5 lines)
  return prev
```

Explanatory Mode example

```
You:  Explain binary search.

Claude:  Binary search works on
sorted arrays.  The key insight
is:  by checking the middle element,
we eliminate half the remaining
candidates each step.

Why sorted?  Because if arr[mid]
is too large, everything to its
right is also too large.  This
guarantee is what makes the
O(log n) complexity possible.

Think of it like a phone book...
```

Demo: Ask Claude to teach you a concept in Learning or Explanatory mode

Gemini Guided Learning

Launched: August 2025 | **Built on:** LearnLM — Google's education-specific AI

How it works:

- Socratic method + step-by-step guidance
- Won't just give you answers
- Asks questions to ensure you learned the concept
- LearnLM integrated into Gemini 2.5

Key features:

- **Canvas quizzes** — personalized quizzes for any subject
- **Video Overviews** — turn sources into educational videos
- **SAT prep** — free practice tests (Princeton Review)
- Free for educational institutions

LearnLM stats

- Only **0.1%** factual error rate
- Students **5.5 pp** more likely to solve novel problems
- Preferred by educators over competitors

Demo idea

Open Gemini → Guided Learning → ask about a concept. Notice the Socratic style.

Gemini: More Education Features

Gemini for Students:

- Integration with Google Classroom, Docs, Slides
- Ask questions while reading documents
- Get help with assignments
- Research topics within your workflow

For teachers:

- Draft assignments in Classroom
- Summarize student progress
- Record/attach audio & video for feedback

Example interaction

```
You: What causes ocean tides?

Gemini: Good question! Before I
explain, let me check what you
already know. Can you think of
what celestial body might have the
biggest influence on tides?

You: The moon?

Gemini: Right! Now, why do you
think the moon specifically?
What property does it have that
could affect Earth's oceans?
```

Side-by-Side Comparison

	ChatGPT	Claude	Gemini
Mode name	Study Mode	Learning / Explanatory	Guided Learning
Approach	Refuses answers, quizzes	Socratic + `#TODO` inserts	Socratic questioning
Best for	Math/science, general study	Coding, technical concepts	Google ecosystem, quizzes
Unique	70+ interactive topics	Pair-programming style	LearnLM, Video Overviews
Free tier	Yes	Yes	Yes (free for edu)
Ecosystem	Canvas, Custom GPTs	Artifacts, Projects	Classroom, Docs, Slides

Key insight: all three are designed to **guide**, not **give answers**.
Try the same question in all three — you'll notice different tutoring styles.

Same Question, Three Styles

Question: *"Explain how recursion works in programming."*

Before I explain, tell me: have you heard of a function calling itself? What do you think would happen if a function called itself without stopping?

→ Tests your understanding first

Claude Learning Mode

Let's learn by building. Here's a skeleton for factorial:

```
def factorial(n):
  # TODO: Write the base
  # case and recursive call
```

→ Makes you write the code

Gemini Guided

Great topic! Think of Russian nesting dolls — each doll contains a smaller one. What would be the "smallest doll" in a recursive function?

→ Uses analogies, then guides

Optional demo: ask the same question to all three and compare live

NotebookLM: Your Personal Study Assistant

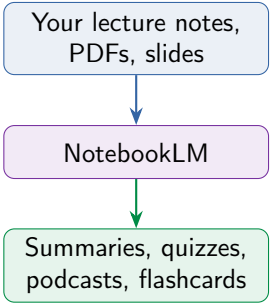
What: Upload YOUR materials → AI assistant grounded in YOUR sources

How it's different from chatbots:

- Regular LLMs answer from training data (generic internet)
- NotebookLM answers from **your uploaded documents**
- Source-grounded = fewer hallucinations
- Every answer references your materials

What you can upload:

- PDFs, Google Docs, Slides
- Lecture notes, textbook chapters
- Research papers, YouTube videos, audio



NotebookLM: Key Features

Audio Overviews

AI generates a **podcast-style discussion** of your materials. Two AI hosts discuss your content. Listen on commute, exercising, etc.

Interactive Mode

"Raise your hand" during the podcast. AI hosts stop, answer your question, then continue.

Flashcards & Quizzes

Auto-generated from your documents. Test yourself on **your actual course content**.

Learning Guide

Structured, deeper exploration of your materials. Beyond summaries — builds understanding.

Price

Free for all education users.

Demo: upload a lecture PDF, generate an Audio Overview, try Interactive Mode

Perplexity AI: Research with Citations

What: AI search engine — every claim comes with a source

Why it matters for students:

- Unlike ChatGPT/Claude: **cites sources** for every claim
- Better than Google for research questions
- Synthesizes from multiple sources
- Academic-quality research in seconds

Key features:

- **Study Mode** — flashcards, summaries, quizzes
- **Research Mode** — in-depth academic analysis
- **Expanded Citations** — up to 10× more sources
- Wiley partnership for academic content

Free Pro for students

- **12 months free** with edu email
- Referral program: earn up to 24 free months
- Includes all premium features

Caveat

Best citation accuracy among AI search engines, but still **37% error rate**. Always verify!

Demo: compare Google search vs. Perplexity for a research question

Khanmigo: Purpose-Built AI Tutor

What: Khan Academy's AI tutor — designed for education from day one

How it works:

- Socratic method: guides, **never gives direct answers**
- Limitless patience — guides until you get it
- Built on GPT-4, constrained for education
- Integrated with Khan Academy's content

Features:

- **Image upload** — scan handwritten math
- Writing coach, debate partner
- Career coaching, speech support

Availability

- \$4/month or \$44/year
- Free via many school districts
- 10+ languages: English, Spanish, Russian, Arabic, Hindi, Chinese...

For teachers:

- Standards-aligned lesson plans
- Student progress summaries
- Learning objectives, rubrics, quizzes

Math & Science Tools

Wolfram Alpha

- Computational knowledge engine
- Step-by-step solutions
- Advanced math, science, engineering
- Curated data + algorithms
- Gold standard for computation

Best for: advanced problems

Photomath

- Mobile app — scan and solve
- Handwritten or printed math
- Instant step-by-step solutions
- Arithmetic through calculus

Best for: quick help on the go

Symbolab

- Step-by-step algebra & calculus
- Practice problems with hints
- Interactive graphing
- Free tier available

Best for: learning step by step

Pro tip: combine ChatGPT + Wolfram Alpha — ChatGPT guides the *process*, Wolfram verifies the *computation*.

AI for Writing & Research

Grammarly AI

- Grammar, spelling, punctuation
- Style & tone suggestions
- Plagiarism detection
- Works in browser, Word, Google Docs
- Free tier available

Best for: polishing your writing

Notion AI

- Organize notes & knowledge
- Summarize long documents
- Generate content in your workspace
- Q&A over your notes
- Student plan available

Best for: organizing study materials

Gamma AI

- AI presentation creation
- Text → beautiful slides in minutes
- No design skills needed
- Websites & documents too
- Free tier available

Best for: creating presentations fast

AI for Coding

GitHub Copilot

- AI pair programmer in your editor
- Code autocomplete + chat
- Explains code, suggests fixes
- **Free for students** (GitHub Education)
- Works in VS Code, JetBrains, etc.

Replit AI

- Browser-based coding environment
- AI assistance built in
- No setup needed — great for beginners

Other coding assistants

- **Claude Code** — terminal-based AI agent
- **Cursor** — AI-first code editor
- **AskCodi** — debug & explain (free)
- **ChatGPT / Claude / Gemini** — paste code, ask questions

Learning tip

Don't just accept generated code. Use **Learning Mode** or ask: *"Explain this code line by line."*

AI for Language Learning

Dedicated tools

- **Duolingo** — gamified, AI-adaptive, roleplay conversations
- **Speak** — AI conversation partner
- **ELSA Speak** — pronunciation feedback

Why LLMs are game-changers:

- Free-form conversation in any language
- Correct mistakes with explanations
- Adapt to your level automatically
- Available 24/7

Using LLMs for languages

- **Voice modes** — speak and listen
- Practice real conversations
- Grammar explanations in your language
- Role-play: ordering food, interviews...

Try this prompt

"Let's talk in French. I'm B1. Correct my mistakes gently and explain why."

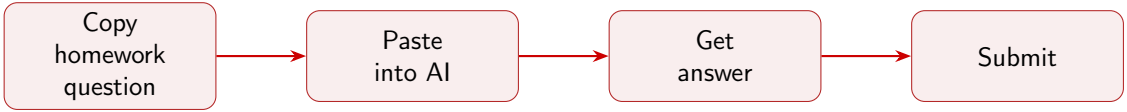
The AI Learning Toolkit

What do you need? → Use this tool

Task	Best Tool	Why
Research with sources	Perplexity AI	Citations for every claim
Study from your notes	NotebookLM	Grounded in your materials
Understand a concept	ChatGPT / Claude / Gemini	Learning/education modes
Solve math step-by-step	Wolfram Alpha / Photomath	Computational accuracy
Write code	GitHub Copilot / Claude	Free for students, explains code
Improve your writing	Grammarly AI	Grammar, style, tone
Create a presentation	Gamma AI	Text → slides in minutes
Practice a language	Duolingo + LLM voice	Structured + free-form practice
Organized note-taking	Notion AI	Summarize, search, generate

Best results come from combining tools, not relying on one.

The Wrong Way



What happens

- Zero effort → zero learning
- False confidence on exams
- Fragile knowledge that crumbles under pressure
- Can't explain your own work

The exam problem

- You can't bring AI to the exam
- If you never struggled with the material, you never learned it
- **Struggle is where learning happens**

The Right Way: AI as Tutor, Not Calculator

1. Try first
Attempt the problem

→

2. Get stuck
Identify what you don't get

→

3. Ask AI
to *guide*, not solve

→

4. Do it
yourself with guidance

→

5. Verify
Cross-check with textbook

The key difference

- "Help me understand X" ✓
- "Solve X for me" ×
- You do the thinking — AI guides
- Learning modes enforce this!

Why this works

- You still struggle (that's learning!)
- AI provides scaffolding, not answers
- You build real understanding
- Knowledge that survives exams

Prompting for Learning (Part 1)

Prompts that maximize learning:

Guide, don't solve

- *"Don't give me the answer. Guide me step by step."*
- *"I think the answer is X. Am I right? Where did I go wrong?"*
- *"What's wrong with my approach?"*

Explain & understand

- *"Explain this using an analogy."*
- *"What's the intuition behind [formula]?"*
- *"Why does this work? Not how — why."*

Test yourself

- *"Quiz me on [topic] — start easy, then harder."*
- *"Give me 5 practice problems similar to this."*
- *"I studied X. Check if I really understand."*

Go deeper

- *"What concept am I missing?"*
- *"What are common misconceptions here?"*

Prompting for Learning (Part 2)

More advanced techniques:

Level adjustment

- *"Explain like I'm a complete beginner."*
- *"Now explain at a graduate level."*
- *"Use only concepts from Calculus 1."*

Connections & synthesis

- *"How does [A] relate to [B]?"*
- *"Real-world example of [concept]."*
- *"Compare and contrast [X] and [Y]."*

Role-based learning

- *"Act as a [subject] professor — give me a 5-min lecture on [topic]."*
- *"Be my study partner. Let's challenge each other."*

Exam prep

- *"Study plan for [topic], 2h/day, 5 days."*
- *"Practice exam: 10 questions on ch. 5–8."*
- *"Most likely exam questions on [topic]?"*

Creative LLM Learning Hacks

Unconventional ways to squeeze more learning from every interaction:

Passive language immersion

"Replace 3–5 words in every response with their French equivalent (in parentheses)." Absorb vocabulary **without dedicated study time**.

Daily drip — scheduled learning

Set a daily reminder to prompt the LLM: *"Teach me one new concept about [topic] in 3 sentences."* Micro-learning compounds over weeks.

"Always teach me something new"

"End every response with one surprising fact about what we discussed." Turns every chat into discovery.

Teach-back method

Explain a concept **to the LLM**, then ask: *"Find the holes in my explanation."* Teaching is the best way to learn.

"Debug my thinking"

"Here's my mental model of [X]. Where am I wrong?" Surfaces blind spots you didn't know you had.

Counterfactual exploration

"What if Newton never discovered calculus?" "What if" questions build deeper intuition by exploring **why** things are the way they are.

Combining Tools: A Study Session Workflow

1. Upload lecture notes to **NotebookLM** → listen to Audio Overview on commute
2. Use **Perplexity** to research related concepts with citations
3. Ask **ChatGPT/Claude/Gemini** in learning mode to quiz you
4. Practice problems — verify answers with **Wolfram Alpha**
5. Create summary flashcards in **NotebookLM** for spaced review

Each tool has a strength — combine them for maximum learning.

Hallucinations: AI Makes Stuff Up

What are hallucinations?

- LLMs generate plausible-sounding but **wrong** info
- They don't "know" facts — they predict likely text
- Can invent citations, formulas, historical events
- Sounds confident even when completely wrong

Educational risks:

- Learning wrong formulas or definitions
- Citing non-existent papers
- Building understanding on incorrect foundations
- Perplexity: best accuracy, still **37% error rate**

How to protect yourself

1. **Always verify** with textbook/professor
2. Use **NotebookLM** for source-grounded answers
3. Use **Perplexity** for cited research
4. **Cross-reference** multiple tools
5. Be skeptical of confident answers

Rule of thumb

If AI says something surprising, verify before relying on it.

Academic Integrity

Know your institution's policy

- AI policies vary **widely**
- Some ban all AI, some encourage it
- Some allow for understanding, not submissions
- **Check before using AI for assignments**

Be transparent

- Disclose AI use when submitting work
- Explain *how* you used it
- "Used Claude to understand, wrote solution myself"

The ethical line

- AI to **understand** $\neq$ AI to **cheat**
- Tutor: questions, explanations ✓
- Ghostwriter: generating submissions $\times$
- Learning modes keep you on the right side

Remember

If you can't explain your own work without AI, you didn't learn it — and it will show on the exam.

Privacy & Data

Be careful what you share:

- Don't paste personal/sensitive info
- Don't share credentials or financial data
- Don't upload others' private data
- Be cautious with unpublished research

Data retention policies differ:

- ChatGPT: may train on chats (opt out)
- Claude: doesn't train by default
- Gemini: check settings
- NotebookLM: data stays in your account

Best practices

1. Use **institutional accounts** when available
2. Check opt-out settings for training
3. Don't share what you wouldn't post publicly
4. Use "temporary chat" modes

Institutional offerings:

- ChatGPT Edu — enterprise privacy
- Claude for Education — campus deployments
- Gemini for Education — Workspace protections

ArmBench-LLM: Which Models Speak Armenian?

ArmBench-LLM 1.0 — first comprehensive benchmark for Armenian language tasks. Created by **Metric AI Lab** (April 2026). Fully open-sourced.

7 task categories: text classification, multiple choice QA, grammar correction, space fixing (OCR), summarization, translation, reading comprehension.

Rank	Model	Score	Cost
1	**Gemini 3 Flash**	0.635	$3.28
2	GPT-5.2 Pro	0.617	$160.20
3	Grok 4 Fast	0.604	$2.71
4	Claude 3.7 Sonnet	0.607	$16.49
5	**Qwen 3.5 27B** (open-source)	0.577	free

Key insight

Global rankings $\neq$ Armenian proficiency! Gemini Flash **beats** Gemini Pro on Armenian (opposite of global ranking).

Open-source win

Qwen 3.5 27B beats 600B+ models (GLM-5, Mistral-Large). Runs on a **single GPU**.

ArmBench: Task-Specific Results

Best models by task:

Task	Best model
Translation	Gemini 3 Flash
Summarization	Gemini 3 Flash
Armenian Literature	Gemini 3 Flash
Armenian History	Gemini 3 Pro
Math (university exams)	Grok 4 Fast
Named Entity Recog.	Grok 4 Fast

Cost matters:

- GPT-5.2 Pro costs **50**× more than Gemini Flash for *lower* accuracy
- Armenian tokenizers vary wildly — same text = different cost
- Reasoning models burn tokens on chain-of-thought

Practical recommendations:

Best overall

Gemini 3 Flash — highest score, lowest cost. Best for translation, summarization, literature.

Best budget

Grok 4 Fast — nearly as good, cheapest ($2.71). Best for math, NER.

Best open-source

Qwen 3.5 27B — run locally, no API costs, beats much larger models.

Try it yourself

Leaderboard: `metric-ai-armbench-llm.hf.space`
Search `Metric-AI/ArmBench-LLM-data` on HF

Key Takeaways

1. Use **learning/education modes**, not default modes
2. AI should make you **think more**, not less
3. **Combine tools** — each has different strengths
4. **Always verify** — AI hallucinates
5. **Be transparent** about AI use

Free Access for Students

Tool	Free access	How
ChatGPT	Study Mode on free tier	Sign up at chat.openai.com
Claude	Free tier available	Sign up at claude.ai
Gemini	Free for edu institutions	Via Google Workspace for Education
NotebookLM	Free for everyone	notebooklm.google
Perplexity Pro	12 months free	Edu email at perplexity.ai
GitHub Copilot	Free for students	GitHub Education pack
Wolfram Alpha	Free basic access	wolframalpha.com
Khanmigo	$4/mo (or free via school)	khanmigo.ai
Grammarly	Free tier available	grammarly.com
Gamma AI	Free tier available	gamma.app

Most of these tools are free or have generous student programs.
You have no excuse not to try them!

Resources & Links

The Big Three:

- ChatGPT: `chat.openai.com`
- Claude: `claude.ai`
- Gemini: `gemini.google.com`

Study & Research:

- NotebookLM: `notebooklm.google`
- Perplexity: `perplexity.ai`
- Khanmigo: `khanmigo.ai`
- Wolfram Alpha: `wolframalpha.com`

Coding:

- GitHub Copilot: `github.com/features/copilot`
- GitHub Education: `education.github.com`
- Replit: `replit.com`

Writing & Productivity:

- Grammarly: `grammarly.com`
- Notion AI: `notion.so`
- Gamma AI: `gamma.app`

Questions?

Use AI to learn better — not to learn less.